

AnalystLab Africa Internship Program – Week 3 SQL & Data Querying Project

Chinook Database Analysis

Introduction

Structured data is commonly stored in relational databases, making SQL one of the most important skills for data analysts. This project explores the Chinook Music Store database using SQL to extract, analyze, and interpret business information.

The analysis focuses on customer purchasing behavior, product performance, revenue trends, and customer segmentation. Various SQL concepts were applied, including filtering, aggregation, joins, subqueries, window functions, and query optimization techniques.

Objective

The objectives of this project were to:

- Practice querying relational databases using SQL.
- Apply aggregate functions to summarize business performance.
- Use joins to combine information across multiple tables.
- Analyze customer purchasing patterns and revenue generation.
- Apply advanced SQL concepts such as subqueries and window functions.
- Generate business insights from query results.

Dataset Overview

The Chinook database is a sample digital music store database containing information about customers, invoices, artists, albums, tracks, genres, and sales transactions.

Key tables used in this analysis include:

- Customer
- Invoice
- InvoiceLine
- Track
- Album
- Artist
- Genre

These tables were connected using primary and foreign key relationships to answer business-focused analytical questions.

Analysis 1: Customer Purchase Analysis

Purpose

Identify customers who contribute the most revenue to the business.

Key Findings

- Helena Holý was the highest-spending customer with total purchases of **\$49.62**.
- Richard Cunningham ranked second with **\$47.62**.
- The difference between the highest and fifth-highest spending customers was only **\$4.00**, indicating relatively balanced customer spending.

Business Insight

Revenue is distributed across multiple customers rather than being dependent on a small number of high-value buyers. This reduces customer concentration risk and suggests a stable customer base.

Analysis 2: Top-Selling Music Genres

Purpose

To identify which music genres generate the highest revenue and contribute most to overall sales.

Key Findings

- Rock generated **\$826.65** in revenue, making it the highest-performing genre.
- Latin generated **\$382.14** in revenue and ranked second.
- Metal generated **\$261.36** in revenue and ranked third.
- Rock generated more than double the revenue of Latin.

Business Insight

Rock is the dominant revenue-generating genre in the Chinook Music Store. The significant revenue gap between Rock and other genres suggests strong customer preference for Rock music. The company could prioritize Rock-related promotions, recommendations, and inventory expansion to maximize revenue opportunities.

Analyst Observation

Rock generated **\$444.51** more revenue than Latin and generated more revenue than Latin and Metal combined, highlighting its importance to overall sales performance.

Analysis 3: Top Artists by Revenue

Purpose

To identify the artists that generate the highest revenue and contribute most to music sales.

Key Findings

- **Iron Maiden** generated **\$138.60** in revenue, making it the highest-performing artist.
- **U2** ranked second with **\$105.93** in revenue.
- **Metallica** generated **\$90.09** in revenue.
- **Led Zeppelin** generated **\$86.13** in revenue.
- The television series **Lost** generated **\$81.59** in revenue, making it one of the highest-performing non-music content categories.

Business Insight

A relatively small group of artists contributes a significant share of total revenue. Iron Maiden leads all artists, indicating strong customer demand for their catalog. The strong performance of artists such as U2, Metallica, and Led Zeppelin suggests that classic rock and metal music are major drivers of sales within the Chinook database.

Analyst Observation

The top four revenue-generating artists are all associated with the Rock and Metal genres, which supports the findings from the genre analysis where Rock was identified as the highest-performing genre. This demonstrates a strong relationship between artist popularity and genre performance.

Analysis 4: Monthly Revenue Trend Analysis

Purpose

To analyze how revenue changes over time and identify periods of higher or lower sales performance.

Key Findings

- Monthly revenue remained relatively stable throughout the observed period.
- Most months generated approximately **\$37.62** in revenue.
- The highest monthly revenue was recorded in **January 2022 (\$52.62)**.
- Other strong-performing months included:
 - April 2023 (**\$51.62**)
 - June 2023 (**\$50.62**)
 - November 2025 (**\$49.62**)

- The lowest monthly revenue was recorded in **November 2023 (\$23.76)**.

Business Insight

Revenue performance remained consistent across most months, indicating stable customer purchasing behavior. However, occasional revenue spikes suggest periods of increased customer activity that could be investigated further to identify successful promotions, product releases, or seasonal demand patterns.

Analyst Observation

The difference between the highest and lowest revenue months is:

$$\$52.62 - \$23.76 = \$28.86$$

This indicates that while revenue is generally stable, certain periods experience significantly stronger or weaker sales performance.

Analysis 5: Customer Spending Ranking

Purpose

To rank customers based on their total spending and identify the highest-value customers.

Key Findings

- Helena Holý ranked **1st** with total spending of **\$49.62**.
- Richard Cunningham ranked **2nd** with **\$47.62**.
- Luis Rojas ranked **3rd** with **\$46.62**.
- Hugh O'Reilly and Ladislav Kovács shared **Rank 4** because they spent the same amount (**\$45.62**).
- Puja Srivastava ranked last with total spending of **\$36.64**.

Business Insight

The ranking analysis shows that customer spending is relatively balanced across the customer base. The highest-spending customer spent only **\$12.98** more than the lowest-spending customer, indicating that revenue is not heavily dependent on a small group of customers.

Analyst Observation

The ranking results reveal an important feature of the `RANK()` function:

- Customers with equal spending receive the same rank.

- Hugh O'Reilly and Ladislav Kovács both spent **\$45.62**, so they were both assigned Rank 4.
- The next rank becomes Rank 6 rather than Rank 5.

This behavior helps analysts correctly identify ties in performance rankings.

Analysis 6: Customers Above Average Spending

Purpose

To identify customers whose total spending exceeds the average customer spending level.

Key Findings

- A total of **22 customers** spent more than the overall customer average.
- Helena Holý recorded the highest spending at **\$49.62**.
- Richard Cunningham spent **\$47.62**.
- Luis Rojas spent **\$46.62**.
- Several customers recorded spending above **\$40**, placing them among the most valuable customers in the database.

Business Insight

A relatively small group of customers contributes above-average revenue to the business. These customers represent a high-value customer segment that could be targeted through loyalty programs, personalized promotions, and retention strategies.

Analyst Observation

This query used a **subquery** to calculate the average customer spending and then compared each customer's spending against that benchmark. This approach allows businesses to identify customers who perform above average and focus retention efforts on their most valuable customers.

Analysis 7: Query Optimization

Purpose

To improve database performance and reduce query execution time.

Optimization Techniques Applied

Indexes were created on frequently used columns:

```
CustomerId (Invoice Table)
TrackId (InvoiceLine Table)
```

Business Benefit

Indexes help databases locate records more efficiently, especially when working with large datasets. This improves query performance and reduces processing time for joins, filtering operations, and analytical reporting.

Conclusion

You can end your report with this:

Conclusion

This project demonstrated the practical application of SQL for business analysis using the Chinook Music Store database.

Key findings revealed that:

- Helena Holý was the highest-spending customer with total purchases of **\$49.62**.
- Rock generated the highest revenue (**\$826.65**) among all music genres.
- Iron Maiden was the top-performing artist, generating **\$138.60** in revenue.
- Revenue remained relatively stable over time, with occasional periods of increased sales activity.
- Twenty-two customers spent above the overall customer average, representing a valuable customer segment for targeted business strategies.

The project also applied advanced SQL concepts including joins, aggregate functions, subqueries, window functions, and indexing techniques. These skills are essential for extracting meaningful insights from relational databases and supporting data-driven decision-making.
